Ananya Prabhu

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Summary

Data Engineering and AI graduate with research experience in real-time sensor systems and explainable AI. Built a multi-person radar monitoring pipeline at Tampere University, leading to a first-author paper submitted to IEEE Sensors Letters.

Publications

- **A. Prabhu**, A. Site, T. Lipping. *Human Activity Recognition Using Ceiling-Mounted Radar Sensors.* arXiv preprint arXiv:2610.39628, Oct 2026. Submitted to *IEEE Sensors Letters* (APSCON 2027 track), under review.
- **A. Prabhu**. *Explainable Artificial Intelligence for Emotion Classification: LIME and SHAP Analysis on Labeled Emotion Data.* Bachelor's thesis, Turku University of Applied Sciences, 2026. theseus.fi/handle/10024/927587.

Research & Work Experience

Research Intern - SmartCare Environment Project 2026 - Present
Tampere University, Pori Campus

- Sole developer of an end-to-end real-time detection pipeline for three ceiling-mounted radar sensors: MQTT ingestion, DBSCAN clustering, position smoothing and multi-person tracking (up to 3 people).
- Built a six-view web dashboard (room map, point cloud, zero-Doppler, activity, clinical, replay) covering two research spaces, with posture classification, breathing-rate estimation and gait analysis.
- Designed and led data collection sessions; ran exploratory analysis of radar signatures across activities, which forms the evaluation of the submitted journal paper.

Intern – INSIGHT Lab (Health Tech) Jan 2025 – Mar 2025
Turku University of Applied Sciences

- Cleaned and organised medical imaging datasets into BIDS structure, including metadata collection.
- Extended a data preprocessing pipeline to handle multiple datasets with dynamic partitioning.
- Built a testing workflow for segmentation model training across datasets (Python, Pandas, Docker, Linux).
- Applied federated learning for distributed analysis of brain imaging data across sites.

Projects

- **Wine Glass Detection and Style Transfer(Deep Learning Project):** Three-stage pipeline: custom CNN classifier, fine-tuned YOLOv5s detector, and neural style transfer applied only to detected regions.
- **Federated Learning UI/UX (Capstone, INSIGHT Lab, 2025):** led the design of a prototype interface for federated learning systems used in distributed AI research.
- **Head Motion Tracking in Unity (R&D, 2025):** built a head-tracking app with real-time sensor integration for human–computer interaction research.
- **ETL Pipelines with Apache Airflow (2024):** orchestrated pipelines that load, clean and aggregate power-plant and weather data into SQL; coursework pipelines in PostgreSQL with Kafka streaming.
- **Notiva Productivity Dashboard:** React/TypeScript app with task management, a Pomodoro timer and animated dashboards.
- **Game Lending System:** full-stack Flask and SQLite app for tracking game loans like a library.
- **Freelance E-commerce Website (2022–23):** full-stack Django site with Figma UI designs and test-driven development.

Education

Bachelor of Engineering, ICT – Data Engineering & AI Graduated June 2026
Turku University of Applied Sciences, Finland

Skills

Languages: Python, SQL, MATLAB, JavaScript, TypeScript, C (basic), HTML, CSS
AI & Analytics: machine learning, predictive modelling, feature engineering, data preprocessing, CNNs, YOLOv5 object detection, neural style transfer, federated learning, explainable AI (LIME, SHAP), DBSCAN clustering, EDA, data visualization
Data Engineering: data pipelines, ETL/ETVL, Apache Airflow, distributed systems, PostgreSQL, SQLite, MongoDB (basic), NoSQL, Kafka, MQTT
Sensing & Imaging: multi-sensor radar fusion, real-time tracking and dashboards, SimpleITK, rigid and affine registration, BIDS
Frameworks & Libraries: TensorFlow/Keras, PyTorch, scikit-learn, Pandas, NumPy, Flask, Django, React
Tools & Platforms: Git, Docker, Linux CLI, Jupyter, Roboflow, AWS, virtualization, Power BI, Figma, Unity.